

SEVERODVINSK, Russian Federation

WMO: 225460

Lat: 64.5553N

Lon: 39.7656E

Elev: 26

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-9.8	-23.1	1.5	-17.2	-15.2	2.4	-9.8	28.7	28.3	25.7	29.2	4.7	100	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	79.2	63.2	75.2	62.7	71.8	61.4	66.1	74.7	64.5	73.0	62.8	69.6	5.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	86.5	69.3	61.2	80.7	67.9	59.5	76.0	66.5	30.9	74.4	29.6	73.1	28.4	70.1	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.1	16.7	14.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	13.1	21.2	24.4	33.6	47.6	54.6	61.4	58.7	49.5	37.5	26.9	20.3
	DBStd	18.30	14.08	9.90	10.59	7.18	9.43	7.91	6.98	6.36	6.07	6.26	9.33	11.12
	HDD50	5534	1145	806	794	493	164	40	4	4	81	389	693	921
	HDD65	10118	1610	1226	1259	941	543	322	157	212	466	853	1143	1386
	CDD50	967	0	0	0	3	90	178	356	274	65	1	0	0
	CDD65	76	0	0	0	0	4	11	44	17	0	0	0	0
	CDH74	443	0	0	0	0	40	67	254	82	0	0	0	0
	CDH80	64	0	0	0	0	7	6	42	9	0	0	0	0
Wind	WSAvg	7.0	7.3	6.8	7.4	7.0	7.3	7.3	6.2	6.2	6.5	7.6	7.4	7.3
Precipitation	PrecAvg	29.4	2.6	1.9	1.8	1.4	2.0	2.6	2.8	3.1	2.5	2.9	3.0	2.8
	PrecMax	38.9	4.8	3.9	3.1	2.8	3.2	4.3	5.0	5.7	4.4	5.9	4.4	4.6
	PrecMin	24.0	0.7	0.5	0.6	0.3	0.6	0.9	0.8	1.4	1.3	1.0	1.1	1.0
	PrecStd	3.1	0.8	0.9	0.6	0.5	0.6	0.8	1.2	1.1	0.8	1.1	0.7	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	38.6	45.0	61.4	79.4	80.4	84.3	81.2	70.1	53.9	42.8	37.0
		MCWB	33.8	36.0	37.9	45.4	58.6	63.2	65.3	65.7	62.5	49.9	40.6	35.1
	2%	DB	34.0	36.7	40.4	50.9	71.8	75.1	80.3	76.0	63.8	50.5	40.6	35.6
		MCWB	32.7	34.3	35.1	41.4	56.1	61.1	64.0	64.8	57.0	47.1	39.1	33.8
	5%	DB	32.2	35.0	38.1	46.4	66.8	71.5	76.5	72.2	60.9	47.8	38.6	34.2
		MCWB	30.9	33.1	34.1	39.2	53.7	59.0	64.0	62.9	54.5	45.5	37.2	32.9
	10%	DB	30.1	33.6	36.5	43.6	62.9	67.2	72.6	68.8	58.3	45.6	36.7	32.9
		MCWB	28.9	32.1	33.3	38.2	51.4	57.2	62.9	61.4	53.8	43.4	35.3	31.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	36.6	38.5	45.6	61.6	66.5	69.1	67.7	62.4	50.8	41.2	35.7
		MCDB	35.2	38.0	44.0	59.1	72.6	78.5	76.7	74.9	69.3	53.1	42.4	36.7
	2%	WB	32.8	34.5	35.9	42.7	57.7	63.0	66.3	65.9	58.8	47.6	39.3	34.2
		MCDB	33.7	35.9	39.2	47.1	68.6	70.2	75.1	73.2	62.2	49.4	40.5	35.1
	5%	WB	30.8	33.3	34.6	40.5	55.4	60.6	65.0	64.0	56.4	45.6	37.2	33.1
		MCDB	31.9	34.7	37.4	45.6	64.6	68.1	73.9	70.5	59.5	47.8	38.4	34.0
	10%	WB	28.9	32.2	33.6	38.5	52.8	58.4	63.5	62.0	53.8	43.7	35.5	31.8
		MCDB	30.0	33.8	36.1	42.6	61.1	66.3	71.4	67.7	57.7	45.4	36.5	32.9
Mean Daily Temperature Range	5% DB	MDBR	9.6	8.6	10.0	10.9	13.2	11.6	11.5	10.8	9.7	6.0	6.4	8.0
		MCDBR	7.9	6.2	8.7	14.9	21.7	17.9	18.0	17.1	14.6	9.1	5.5	6.9
		MCWBR	8.0	5.5	6.1	9.0	10.6	8.1	7.5	8.7	8.6	7.3	5.5	6.7
	5% WB	MCDBR	8.0	6.2	8.3	13.1	19.0	15.4	15.0	13.5	12.3	8.4	5.7	6.5
		MCWBR	8.2	6.0	6.4	8.3	10.9	8.7	7.8	8.5	8.5	7.3	5.7	6.6
Clear-Sky Solar Irradiance	taub		0.226	0.286	0.314	0.351	0.357	0.350	0.387	0.364	0.337	0.311	0.351	N/A
	taud		1.951	2.078	2.120	2.222	2.418	2.479	2.396	2.441	2.498	2.340	2.533	N/A
	Ebn at Noon		93	187	238	259	270	275	259	255	238	185	89	N/A
	Edh at Noon		8	20	30	34	31	30	32	28	22	16	7	N/A
All-Sky Solar Radiation	RadAvg		39	197	633	1146	1505	1729	1618	1116	614	220	49	13
	RadStd		4	38	81	108	138	152	166	129	69	32	9	1

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page